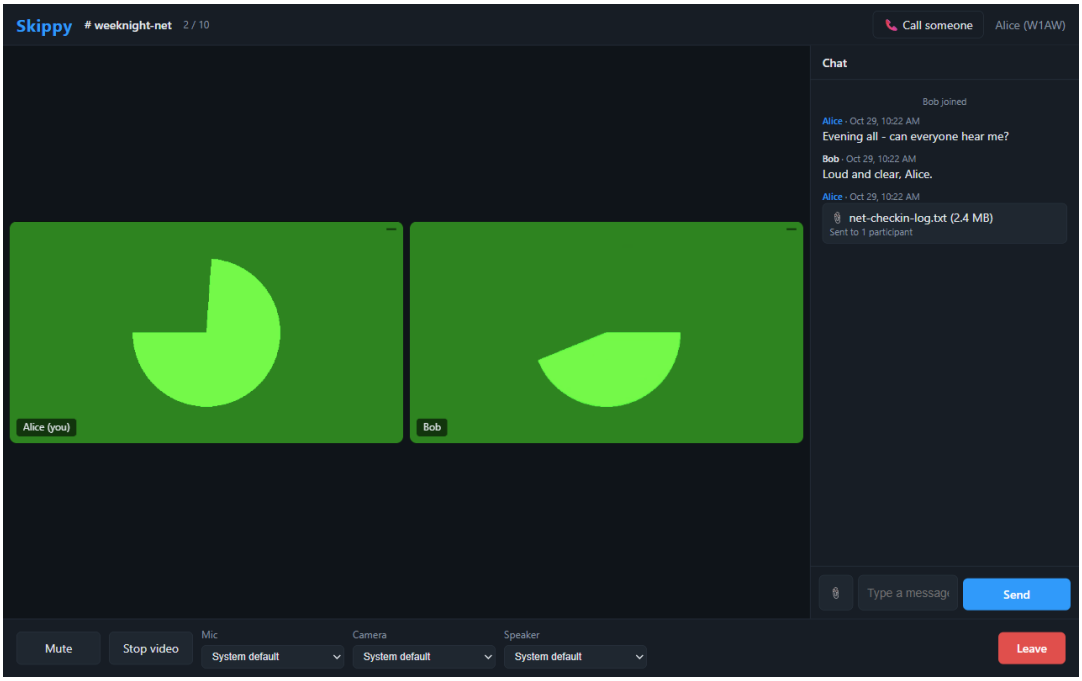
Skippy User Guide

Group video calls for up to 10 people - version 1.10.0

Skippy is a desktop video-conferencing program in the style of Skype. Up to **10 people** can share one call with audio and video, a text chat that keeps its history, and file transfer. You can also **call one specific person** at any time for a private two-person call. Everyone signs in with an account, and an optional authenticator-app code protects it.



A room with two people. Video tiles on the left, chat and file transfer on the right, controls and device pickers along the bottom.

How Skippy fits together

There are two parts. The **Skippy app** is what you install and run on your computer. The **Skippy server** is a small program that one person in your group runs. It keeps the accounts and the chat history, shows who is online, rings people and helps your computers find each other. **Audio, video and files go directly between the participants**, not through the server.

- **Joining an existing group:** install the app and enter the server address your host gives you (step 3 below). That is all.
- **Hosting a group:** also run the server on one computer that stays on while people are using Skippy.

Installing

1. **The app (Windows).** Download *Skippy-Setup-<version>.exe* from the project's Releases page (github.com/lrizio/Skippy/releases) and run it. Windows SmartScreen may say *unknown publisher* because the installer is not code-signed: click **More info**, then **Run anyway**. The installer adds Start-menu and Desktop shortcuts.
2. **The server (the host only).** Install Node.js 22.13 or newer from nodejs.org. Download *Skippy-Server-<version>.zip* from the same page, unzip it and open a command window in that folder:

```
npm install
node server.js
```

On Windows you can instead double-click *Start Skippy Server.bat* after the first *npm install*. The server listens on port **8470** and stores its data in *skippy.db* beside it. If Windows Firewall asks about Node.js, allow it on **private networks**.

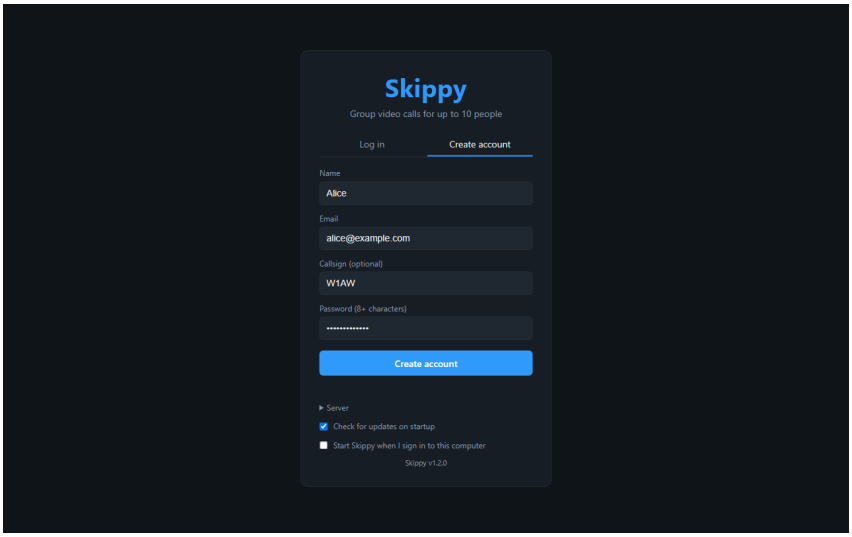
3. Tell the app where the server is. On the login screen, expand **Server** and type the address, for example *ws://192.168.1.20:8470* (the host's computer on your network). The address is remembered. Everyone in a group must use the same server.

Keep the server running. The server window must stay open - **closing it stops the server** and nobody can log in or call. To avoid this, double-click *Run Skippy Server in Background.bat* instead (run *npm install* first). The server then runs hidden, starts each time you sign in to Windows and restarts itself if it ever stops; messages are written to *server.log*. *Stop Skippy Server Background.bat* switches it off again.

Can everyone reach the server? On one home network, yes. Across the internet the host needs to forward port 8470 on the router, or the group can use a VPN such as Tailscale. For use over the internet, put the server behind a TLS (https) front end and use an address starting *wss://*, otherwise passwords travel unencrypted.

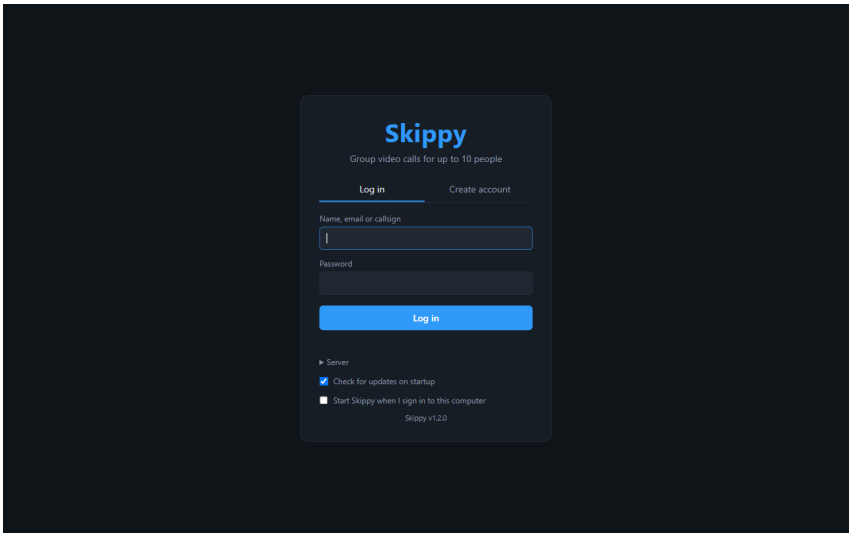
Creating an account and logging in

The first time you run Skippy you must create an account. After that you just log in.



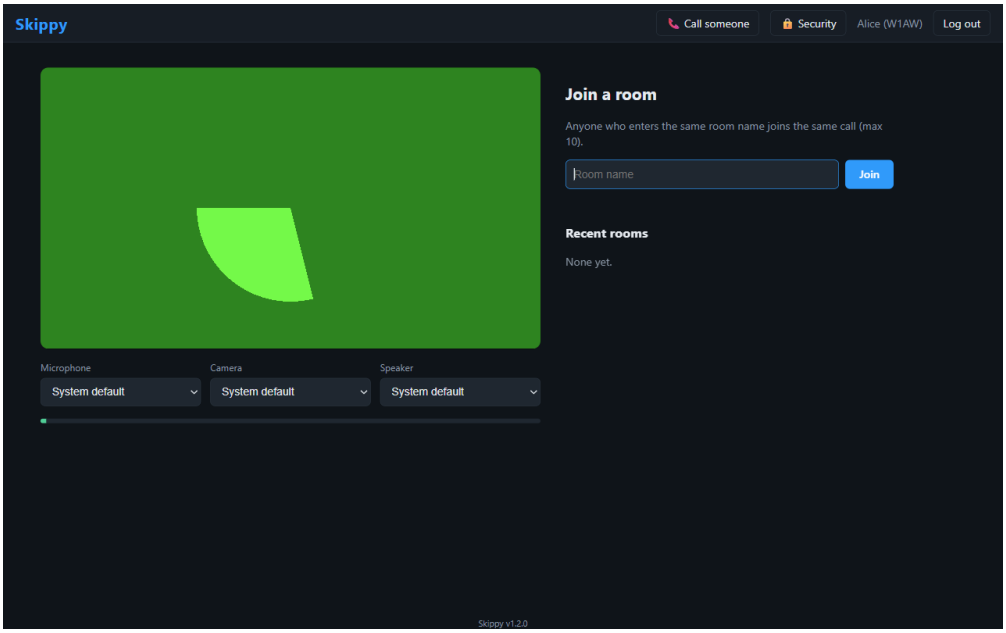
Create account: your name, email, an optional callsign and a password.

- **Name** is what other people see. It must be unique. **Email** is also unique. **Callsign** is optional (letters and digits, e.g. W1AW) and is shown beside your name. The password needs at least 8 characters.
- To log in you can type your **name, email or callsign** - any one of them - and your password.
- Skippy **remembers the last login**: the next time you start the app it signs you straight in as the same person.
- To use a different account, press **Log out** (top right of the lobby). The login screen returns with the last name already filled in; type another name to log in as someone else.



The login screen. Server, update-check and start-at-sign-in options are underneath.

The lobby: choosing your devices



The lobby. Your camera preview and microphone level are on the left; joining a room is on the right.

- Pick your **Microphone**, **Camera** and **Speaker** from the three lists. The preview and the green level bar update at once so you can check them before anyone sees or hears you.
- Your choices are remembered. You can change them again during a call (see *In a call*).
- If Windows asks for permission to use the camera or microphone, allow it. If you have no camera you can still join with audio only.
- A camera or microphone plugged in later appears in the lists automatically.

Joining a room

Type a **room name** and press **Join**. Anyone who types the same name joins the same call, so agree a name with your group (for example *weeknight-net*). Names are not case sensitive. A room holds **10 people**; an eleventh person is told the room is full. Rooms you have used appear under **Recent rooms** for one-click joining. **Rooms in use** lists every named room that has someone in it right now, with a head count - click one to join. (Private direct calls are never listed.)

In a call

Control	What it does
Mute / Unmute	Stops or restarts your microphone. Calls start with your microphone **on**; tick **Join calls with my microphone muted** in the lobby if you would rather start muted. Every video window shows a small microphone symbol in its top-left corner: a microphone when that person can be heard, a crossed-out speaker (red) when they are muted.
Call clock	Next to the head count, shows how long you have been connected to this call.
Stop video / Start video	Turns your camera off or on. Others see your initial on a grey tile while it is off.
Share screen	Pick a whole screen or a single window from the list and everyone in the call sees it instead of your camera. The button becomes **Stop sharing**; press it (or stop from the operating system) and your camera comes back. Audio from the screen is not shared.
Hide chat / Show chat	Turns the chat panel off to give the video more room (the button counts messages that arrived while it was hidden). Drag the thin bar between the video and the chat to make the chat wider or narrower. Skippy remembers both.
Reset layout	Tiles all the video windows neatly again after you have moved or resized them. **Double-click** a window to fill the call area with it; double-click again to go back.
Full screen button	Hover over any video window (including someone's shared screen) and press the small ■ button in its corner to fill the whole screen with it. Press **Esc** (or the button again) to go back. While you share your own screen your tile shows a placeholder instead of your screen, which avoids the endless picture-in-picture mirror effect.
Devices menu	Choose the microphone, camera and speaker without leaving the call. All the buttons are in the title bar at the top.
Leave	Leaves the call and returns to the lobby.
Call someone / **Add someone**	Opens the list of people online so you can ring one of them (see *Calling someone directly*). Inside a call it adds them to that call.
Moving and resizing windows	Every video window is a free-floating window. **Drag it** to move it (for example to put one person above another) and **drag any edge or corner** to resize it, like an ordinary window. Click a window to bring it to the front. The layout lasts until you leave the call or press **Reset layout**.
Header (e.g. *2 / 10*)	How many people are in the room out of the maximum.

Each person has a tile that shows the connection state while it is being set up. Skippy lowers the video quality automatically as more people join (each person's video goes to every other person), so a crowded call looks a little softer than a two-person one. If your internet connection drops, a *Reconnecting* notice appears and Skippy rejoins by itself.

Chat and history

- Type in the box at the bottom right and press **Enter** or **Send**. Everyone in the room sees it at once.
- **Chat history is kept.** Messages are saved on the server per room. When you (or anyone) join again later, the most recent 200 messages appear above a divider, followed by a divider and anything new.
- Joins and leaves are shown as small grey notes.

Sending files

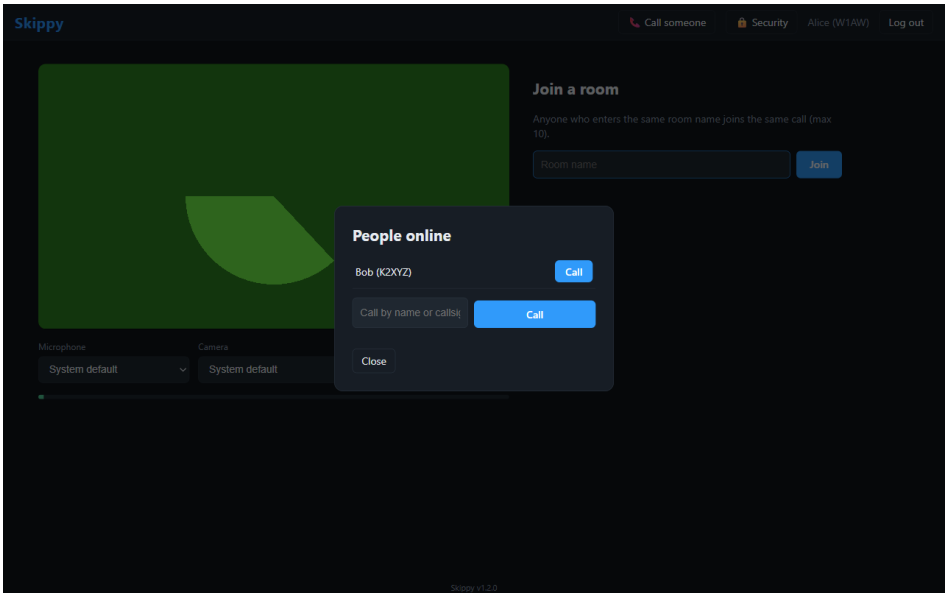
- Click the **paper-clip** button, choose a file and it is sent **straight to everyone in the call** (up to **2 GB**). A progress bar shows on the chat entry; it says *Sent to N participants* when done.
- People receiving it see *Receiving...* and then **Save**. Click Save and choose where to put the file. Received files are written to disk as they arrive, so large files do not use up memory.
- A file is only delivered to the people **in the room when it is sent**. The chat history remembers the file's name and size, but people who join later cannot download it - send it again if needed.

Very large files take as long as your internet upload allows (in a room of several people the file is sent to each person). Closing the call or losing the connection during a transfer cancels it; the receiver sees *Transfer interrupted*.

Calling someone directly

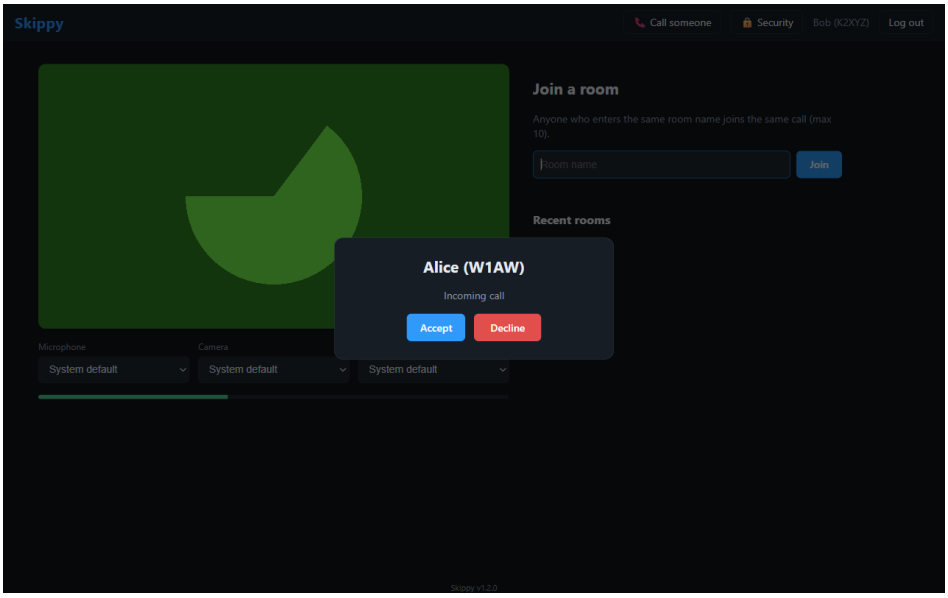
You can call one particular person at any time - from the lobby or while you are in another call - as long as they are logged in to Skippy.

1. Click **Call someone**. The list shows everyone who is online right now.
2. Press **Call** beside a name, or type a **name or callsign** in the box and press Call.
3. Your screen shows *Calling ...* with a **Cancel** button and plays a ring-back tone.



The people-online list.

The other person's Skippy rings, shows who is calling and, if the window was hidden in the system tray, brings it to the front:



The incoming-call dialog on the other person's screen.

- **Accept** opens a private call between the two of you (the header shows a phone icon and their name). If you were in another room, you leave it.
- **Decline** tells the caller. If nobody answers within **45 seconds** the caller is told they did not answer.
- Your private chat with that person has its own history, so the conversation continues next time you call.
- People who are logged out cannot be rung: the caller is told they are not online.

Adding someone to a call. While you are in a call (a room or a direct call), the button at the top reads **Add someone**. Choose a person and they are rung; if they accept they join your current call and you stay where you

are. A direct call becomes a **Group call** (up to 10 people). People added to a private call do not see the chat from before they joined.

Running in the background

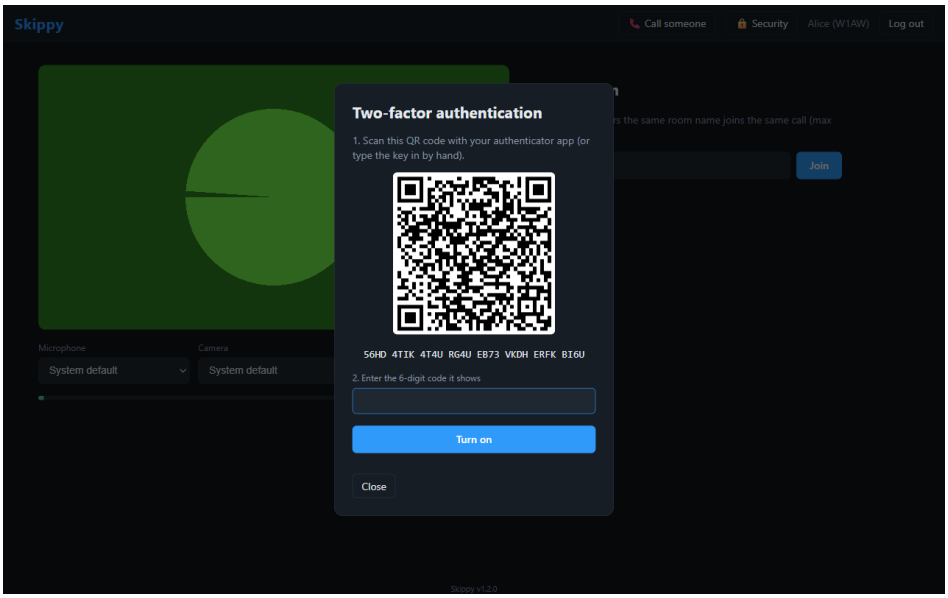
- Closing the Skippy window does **not** quit it. It hides in the **system tray** (the notification area at the bottom right of the taskbar) so it can still ring you. Click the tray icon to open it again.
- To really quit, right-click the tray icon and choose **Quit Skippy**.
- **Logging out** also sends Skippy to the tray. While you are logged out nobody can call you - open it from the tray and log in again when you want to be reachable.
- To have Skippy start (hidden) whenever you sign in to your computer, tick **Start Skippy when I sign in to this computer** on the login screen or in the tray menu.

Two-factor authentication (optional)

Two-factor authentication adds a second step when you log in: a 6-digit code from an authenticator app on your phone. It is optional and set per person.

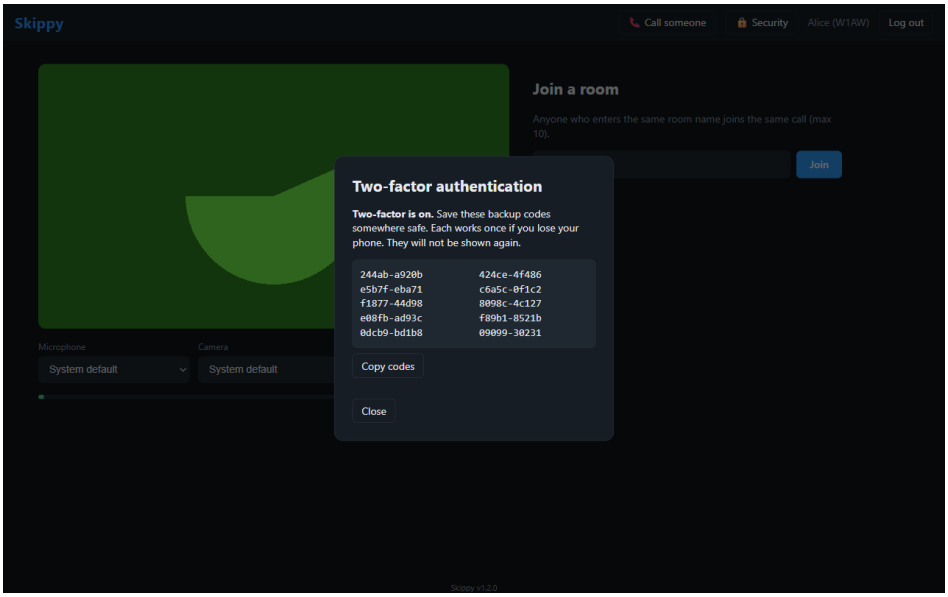
Turning it on

1. Install an authenticator app if you do not have one (Google Authenticator, Microsoft Authenticator, Authy, 1Password and others all work).
2. In the lobby click **Security**, then **Turn on**.
3. Scan the QR code with the app (or type the key shown under it). The app starts showing a 6-digit code.
4. Type the current code into Skippy and press **Turn on**.



Setup: scan the QR code, then enter the code the app shows.

Save your backup codes. Skippy then shows **10 one-time backup codes**. Copy them somewhere safe. Each works once if you lose your phone, and they are **not shown again**.



The backup codes, shown once.

Logging in afterwards. Enter your name and password as usual; Skippy then asks for the authentication code (or a backup code). Starting the app and being signed in automatically does not ask again - the code is needed when you log in fresh, for example after logging out or on a new computer. To turn it off, open **Security** and enter your password and a current code.

> **Lost your phone and your backup codes?** Ask the person who runs your server. They can switch two-factor off for your account by running *node reset-totp.js your-name* in the server folder, after which you can set it up again.

Forgot your password?

When you create an account Skippy shows a **recovery key**, such as *ABCD-EFGH-JKLM-NPQR*. Keep it in a password manager or on paper. If you forget your password, click **Forgot password?** on the login screen, enter your name and the key, and choose a new password. Your other devices are signed out and you are given a new recovery key (the old one stops working). If you use two-factor, you still need your code to finish logging in. Accounts made before this feature can create a key under **Security**.

> **Lost the recovery key too?** Ask the person who runs your server to run *node reset-password.js your-name* in the server folder. It prints a one-time code, good for one hour, that you type in place of the key under **Forgot password?**.

Updates and version

The version number is in the window's title bar and at the bottom of the login and lobby screens. When Skippy starts it checks whether a newer version has been published, either by your server or on the project's GitHub Releases page. If there is one, a banner appears with a **Download** button. To switch the check off, untick **Check for updates on startup** on the login screen. Nothing is downloaded or installed automatically.

Limits and good to know

Item	Limit / behaviour
People per room	10. An eleventh person is told the room is full.
File size	2 GB per file.
Chat	Up to 4000 characters per message; the last 200 messages are shown when you join.
Calling	Only people who are logged in. Rings for 45 seconds.
Same account twice	Joining a room from a second window with the same account moves you there and drops the first.
Account recovery	There is no self-service password reset. Ask the server's host if you forget your password.
Privacy	Audio, video and files are encrypted between participants. The server holds accounts, chat text and file names - not your media or file contents.

Troubleshooting

Symptom	What to check
"Could not reach the server"	Is the server running? Is the address in **Server** right (*ws://host:8470*)? On the same network, check the host's Windows Firewall allows Node.js. Across the internet, is port 8470 forwarded, or are you both on the same VPN?
Everyone shows "connecting" but never connects	Strict routers or firewalls can block direct connections. Try a VPN such as Tailscale, or ask the host about adding a TURN relay (see the project README).
Others cannot hear me	Is the button showing **Unmute** (you are muted)? Is the right microphone chosen? The level bar in the lobby should move when you speak.
No camera picture	Is another program using the camera? Choose it again from the Camera list, or check Windows *Privacy & security > Camera* allows desktop apps.
I cannot hear anyone	Check the Speaker list and your computer's volume. If you switched speakers, pick it again.
Video is blurry with many people	Normal: quality is lowered automatically as the room fills.
"Room is full"	The room already has 10 people.
"Invalid login or password"	Check the name/email/callsign and password. After many wrong tries in a minute you are asked to wait.
Cannot call someone	They must be logged in (shown in the list). If they are logged out or quit the app, you will be told they are not online.
Reconnecting banner keeps appearing	Your connection to the server is unstable, or the server was restarted. Skippy will rejoin when it can.
Windows says "unknown publisher" at install	The installer is not code-signed. Choose **More info** then **Run anyway**.